Solutions

1.

When all rays just reach the sphere, the edge ray is tangent at (Fig.); geometry gives

Fig.

Hence

If the source is over from , some rays miss.

The whole beam reaches the cap if source - distance obeys

For , a ray at to the axis: Fig. . is on the axis; dashed and form a sphere. and are incidence/refraction angles; intersection: . : - axis angle; : refracted ray - axis angle.

The sine rule for gives

The refraction law gives

Fig.

Geometry gives

If its backward extension meets the axis: Fig. . Set the sign of as follows: a ray bent towards the axis gives positive . If it bends away (the backward extension meets the axis left of), is negative; equation still holds.

Fig.

Geometry also gives the fibre-entry condition:

may be negative, meaning is below the axis.

and